

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV **COURSE CODE:** DSA0202

COURSE OUTCOME COVERED

CO2: Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 2 Total Marks:20 Annexure A

Class Test

Code of Conduct:

*I JAASIEL J.S (192524046) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	

Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	
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Annexure A

Class Test

Question 1: Image Preprocessing and Feature Extraction

Given:

- Drone captures aerial images of farmland.
- Images contain noise, shadows, and uneven illumination.
- Objective: Identify pest-affected crop regions.

a) Preprocessing Pipeline

Ans:

A suitable preprocessing pipeline is:

Image Acquisition → Gaussian Filter → Illumination Correction → Histogram Equalization (CLAHE) → Image Normalization

Justification:

- **Gaussian Filter:** Removes noise caused by the camera sensor and wind.
- **Illumination Correction:** Reduces the effect of shadows and uneven sunlight.
- **CLAHE:** Enhances contrast and makes pest-affected regions more visible.
- **Image Normalization:** Maintains uniform pixel intensity for better analysis.

This preprocessing improves image quality and helps in accurate pest detection.

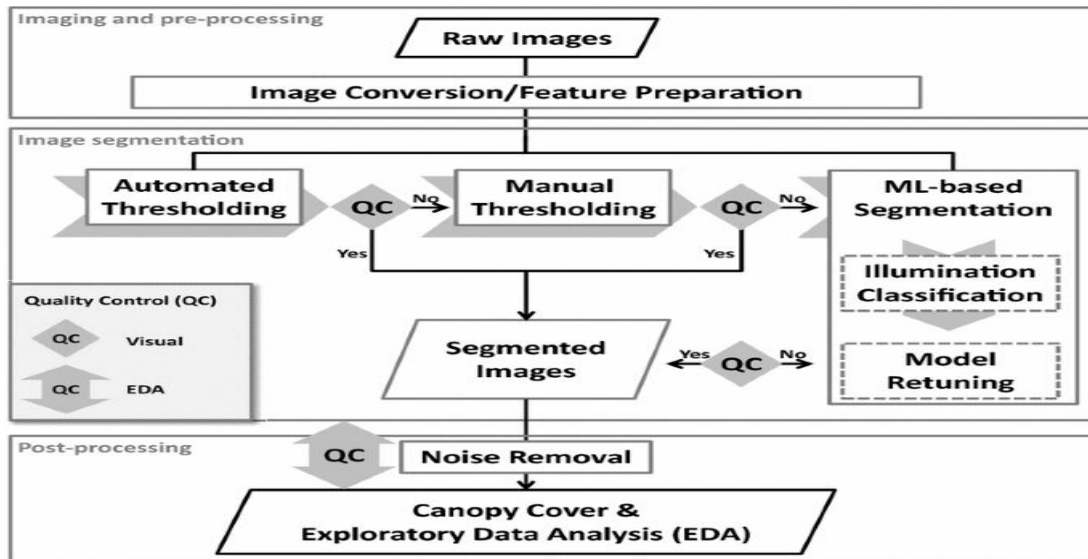


Figure 1: Preprocessing pipeline for improving drone-captured farmland images.

b) Feature Extraction

Ans: A suitable feature extraction method is **SIFT (Scale-Invariant Feature Transform)**.

SIFT detects important texture features and is robust to changes in image scale and rotation. Since pest-affected crops have different texture patterns from healthy crops, SIFT can identify these changes effectively.

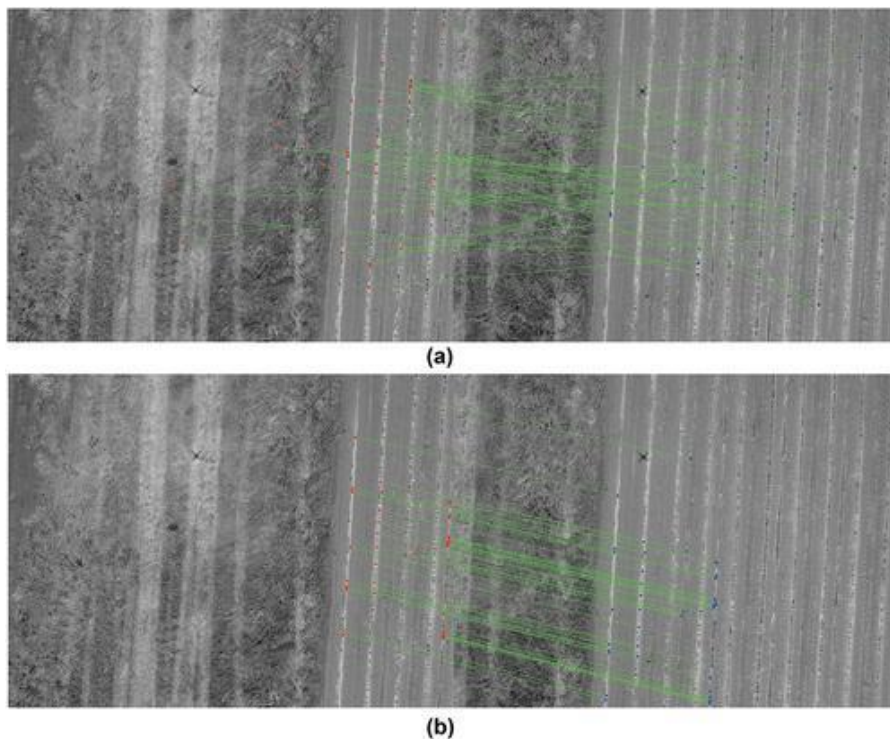


Figure 2: SIFT feature extraction detecting texture changes in crop regions.

c) Rotation and Scale Variations

Ans: Drone images may be captured from different heights and angles, causing changes in image size and orientation. These variations can affect feature detection and matching.

To overcome this problem, **SIFT** or **SURF** can be used because they are **scale-invariant** and **rotation-invariant**, providing reliable feature detection under different viewing conditions.

Question 2: Stereo Vision and Motion Estimation

Given:

- Self-driving car with stereo cameras
- Detect nearby vehicles and pedestrians
- Estimate depth and track moving objects
- Objective: Avoid collisions in real time

a) Stereo Disparity and Depth Estimation

Ans: Stereo cameras capture the same scene from two different positions. The difference in the position of an object in the left and right images is called **disparity**. Using this disparity, the system calculates the distance of the object.

- **Nearby objects** have **larger disparity**, so they appear closer.
- **Distant objects** have **smaller disparity**, so they appear farther away.

This helps the self-driving car estimate the depth of surrounding objects accurately.

Stereo Vision

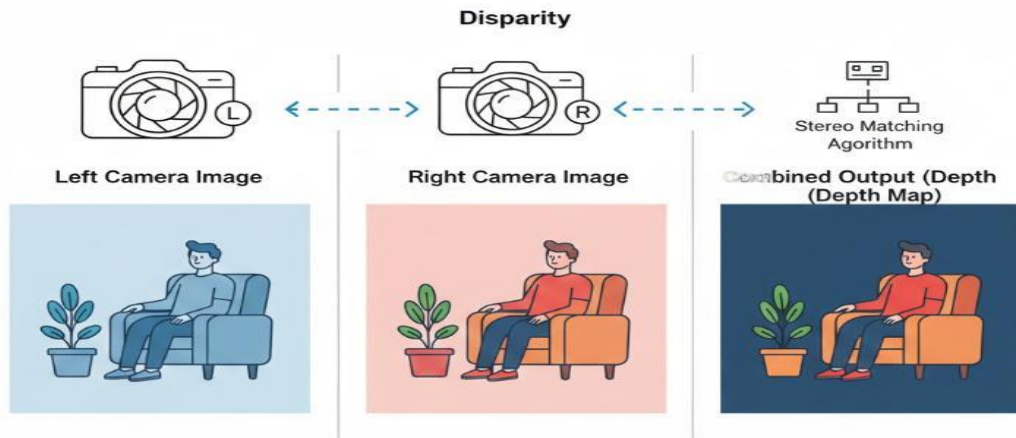


Figure 1: Stereo camera depth estimation using disparity between the left and right images.

b) Optical Flow for Motion Tracking

Ans: Optical flow tracks the movement of a pedestrian by comparing consecutive video frames. If a pedestrian suddenly changes speed or direction, the optical flow vectors also change in direction and magnitude. This helps the system continuously update the pedestrian's position and movement, improving tracking accuracy.

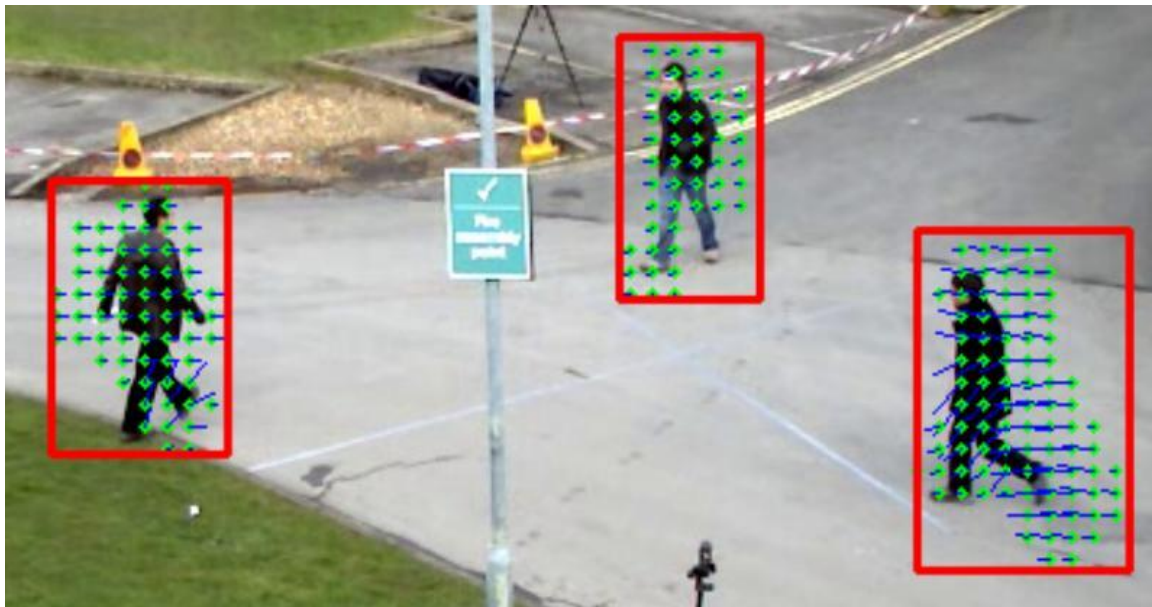


Figure 2: Optical flow tracking a pedestrian between consecutive video frames.

c) Errors and Mitigation Techniques

Ans: Some common sources of error are:

- **Camera misalignment** – causes incorrect depth estimation.
- **Poor lighting** – makes feature detection difficult.
- **Occlusion** – objects become partially hidden.
- **Fast motion** – causes motion blur and incorrect tracking.
- **Textureless regions** – provide very few features for matching.

To reduce these errors:

- **Calibrate the stereo cameras** to avoid misalignment.
 - **Apply image enhancement** to improve visibility in poor lighting.
 - Use **Pyramidal Lucas-Kanade Optical Flow** for better tracking of fast-moving objects.
 - Use **feature-based methods such as SIFT or ORB** to improve matching in difficult regions.
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